

used for finding the precise transmitter frequencies and to set the radio's gain control.

Access address, which is 32-bit long, contains a special value, which means the packet can be received by any Bluetooth device or a unique identifier for a specific connection. Since a Bluetooth controller will be receiving both relevant and irrelevant signals, it must be able to distinguish and accurately pick out the relevant ones.

At the far end, there is the cyclic redundancy check (CRC) that detects cases of transmitted data being unintentionally changed or corrupted due to collisions. It is performed by comparing the CRC calculated by the transmitter with the CRC calculated by the receiver. If they are not the same, then corruption must have occurred, and the packet is discarded.

The message integrity code (MIC), also called a message authentication

code (MAC), is a security feature that aids the detection of data deliberately tampered with (as opposed to accidentally through collisions).

Error correction

There is a variant of the stack's physical layer (PHY) called the LE Coded PHY, which allows Bluetooth to not only detect errors but also correct them in packets up to certain limits. This aids communication at a low signal to noise ratio (SNR).

The further from the transmitter you go, the lower the signal strength and worse the SNR will become, making it harder to keep the BER low. But with forward error correction (FEC), you can work at a low SNR and therefore things continue to be reliable at a much longer range.

Spread spectrum

One of the important ways in which

Bluetooth addresses the reliability challenges is through various ways in which it employs spread-spectrum techniques. So, rather than using a single channel for all packets, they get transmitted across a set of different channels, thus minimising any likelihood of collisions.

Reliable handling of Bluetooth communications

While connecting, devices agree on certain communication parameters, including several that are related to timing, such as how often a radio can be used for servicing this connection. The radio has to be shared potentially across lots of different connections. This particular parameter is called the connection interval.

At the start of the connection interval, devices take turns to transmit and receive with a gap of 150 microseconds in between packets. The precise, agreed timing parameters that control connections provide coordination between the transmit and receive behaviours of the two devices and makes communication reliable.

Connection oriented communication

Spread spectrum in connection oriented communication looks like this: Bluetooth LE divides the 2.4GHz band

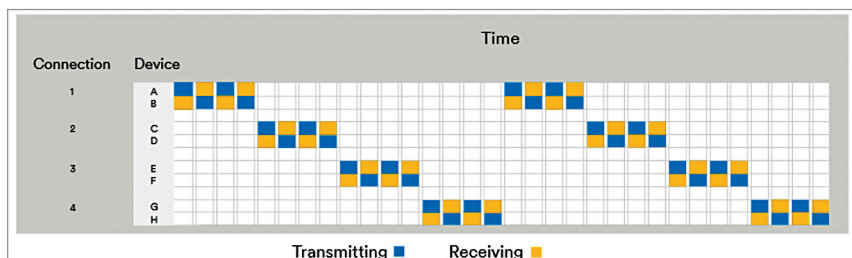


Fig. 7: A simplified illustration of connection use and radio sharing

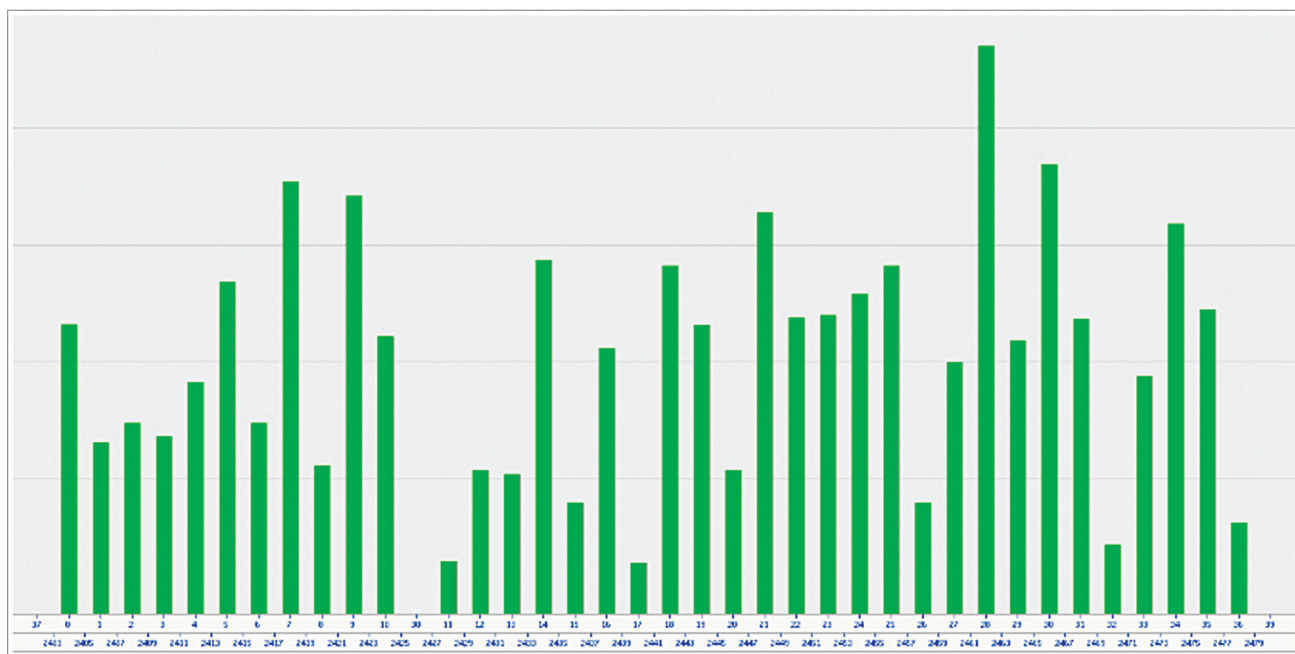


Fig. 8: Adaptive frequency hopping distributing communication across channels